CEKEREK BIOGAS POWER PLANT VALIDATION REPORT

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	Öykü	

Summary:

Brief Description of the Project Activity

"Ruby Canyon Environmental" has conducted the validation of "Cekerek Biogas Power Plant" project activity. The project is a biogas-to-energy that captures biogas from animal manure, which includes dairy cattle, other cattle and chicken, via anaerobic digester and produce clean electricity with one gas engine. This gas engine has an installed capacity of 1.501 MWe. Therefore, the total installed capacity of the project activity is 1.501 MWe.

The project activity is located in Çekerek district, Yozgat Province, Turkey. The project is operated by "Çekerek Enerji ve Gübre Üretim A.Ş.". The start date of the operation (i.e. the start date of the project activity) is 07 December 2021. Since this project is a non-AFOLU project, the renewable crediting period of 7 years, twice renewable for a total of 21 years that is in line with the VCS Standard v4.6. The first crediting period starts on 07/12/2021 and ends on 06/12/2028 (the start and end dates included).

The project comes under Sectoral Scope 01: Energy Industries (renewable/non-renewable sources) (Type I Component) and Sectoral Scope 13: Waste Handling and Disposal (Type III Component). The expected annual emission reduction value is 23,909 tCO_2e/year. Therefore, the total estimated emission reduction for the first crediting period is 167,363

tCO_2e. These GHG emission reductions are claimed as Verified Carbon Units (VCUs) under VCS Standard, version 4.6.

Purpose of the Validation Activity

One of the purpose of this validation activity is to assess if the Project Description Report version 07 dated 10 December 2024 conforms with the requirements of relevant VCS Standard v4.6 criteria which are reviewed in order to confirm that the project design, as documented, is sound and reasonable, and meets the stated requirements and identified criteria. The other purpose is to evaluate the project activity whether it conforms with the monitoring plan and the relevant ex-ante/ex-post parameters.

The Method and Criteria used for the Validation Activity

The validation scope is defined as an independent and objective review of the VCS PD, version 07 dated 10 December 2024, the baseline study of the project, monitoring plan and other relevant documents. The validation was performed based on VCS Program Guide version 4.3 and VCS Standard version 4.6, as well as criteria provided in the CDM methodologies (AMS-III.D, version 21.0 and AMS-I.D, version 18.0) and the methodological tools which reported in the VCS-PD. The validation conducted included three main phases which eventually led to the issuance of this final version of the validation report and opinion, listed below:

- Desk review
- On-site visit and follow-up interviews with the relevant personnel and local stakeholders
- Resolution of outstanding issues and the issuance of final validation report and opinion

Number of Findings Issued during the Validation Activity

During this validation activity, 17 Corrective Action Requests (CARs) and 4 Clarification Requests (CLs) were raised all of which were successfully closed.

Conclusion of the Validation Activity

Ruby Canyon Environmental hereby confirms that the data reported and used in the quantifications of these emission reductions are accurate, complete, transparent, and free of material error and consistent with the project activity's monitoring plan and the requirements set out in the applied methodology, the applied methodological tools, and VCS Standard v4.6. The validation team also confirms that the estimated emission reductions to be claimed during the initial crediting period are a reasonable estimate. RCE issues a positive validation opinion confirming the Project complies with all relevant Methodology requirements and VCS Standard requirements. The validation team did not detect any uncertainties associated with the validation.

Ruby Canyon Environmental also confirms the following based on the results of document review for the period between 07 December 2021 and 06 December 2028:

	Estimated Baseline Emissions or Removals (tCO_2e)	Estimated Project Emissions or Removals (tCO_2e)	Estimated Leakage Emissions (tCO_2e)	Estimated Net GHG Emission Reductions or Removals (tCO_2e)
07/12/2021 – 31/12/2021	1,774	137	0	1,637
01/01/2022 – 31/12/2022	25,902	1,993	0	23,909
01/01/2023 – 31/12/2023	25,902	1,993	0	23,909
01/01/2024 – 31/12/2024	25,902	1,993	0	23,909
01/01/2025 – 31/12/2025	25,902	1,993	0	23,909
01/01/2026 – 31/12/2026	25,902	1,993	0	23,909
01/01/2027 – 31/12/2027	25,902	1,993	0	23,909
01/01/2028 – 06/12/2028	24,128	1,856	0	22,272
TOTAL (07/12/2021 – 06/12/2028)	181,314	13,951	0	167,363

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1 INTRODUCTION

1.1 Objective

Ruby Canyon Environmental has been appointed by “Çekerek Enerji ve Gübre Üretim A.Ş.” to perform the validation of “Cekerek Biogas Power Plant”. The objective of this validation activity is to assess, with objective evidences:

- Whether the implementation and operation of the project activity is compatible with the information provided in the VCS-PD
- Whether all physical features of the project activity indicated in the VCS-PD (e.g. installed technology, monitoring equipment and so on) are in place
- Whether the quantifications of the emission reductions and supporting documents are accurate, complete, transparent, and free of material error and consistent with the project activity’s monitoring plan and the requirements set out in the applied methodology, the applied methodological tools, and VCS Standard v4.6
- Whether the monitoring systems and processes in place are in accordance with the monitoring systems and procedures outlined in the approved methodologies and the monitoring plan in the VCS-PD
- Whether the necessary ex-ante and ex-post data are chosen correctly and monitored in accordance with the applied methodologies, tools and VCS requirements

1.2 Scope and Criteria

The scope of this validation activity which conducted by Ruby Canyon Environmental is the independent and objective review of the project activity and relevant estimated GHG emission reductions. In this regard, the scope is set by:

- The Article 12 of the Kyoto Protocol
- VCS Standard, v4.6, Issued 19 September 2019; updated 21 March 2024 (VCS Standard)
- VCS Program Guide, v4.4, Issued 19 September 2019; updated 29 August 2023
- VCS Validation and Verification Manual, v3.2, 19 October 2016
- CDM Validation and Verification Standard for Project Activities, v3.3
- AMS-III.D: Methane recovery in animal manure management systems, v21.0
- AMS-I.D: Grid connected renewable electricity generation, v18.0

- All associated methodological tools
- Environmental and social issues related to the Sectoral Scope 01: Energy industries (renewable/ non-renewable sources) and Sectoral Scope 13: Waste handling and disposal
- ISO 14064-3:2019 Greenhouse Gases – Part 3: Specification with guidance for the verification validation of greenhouse gas statements

The validation is not meant to provide any consulting towards the client. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the project design. Therefore, Ruby Canyon Environmental can't be held liable by any party for decisions made or not made based on the validation, which will go beyond that purpose.

1.3 Reasonableness of Assumptions

Ruby Canyon Environmental hereby confirms that the reasonableness of assumptions of this validation report is reasonable, with respect to material errors, omissions and misrepresentations. To guarantee this reasonableness all data that is used in the GHG emission reduction calculations have been reviewed without any sampling.

1.4 Summary Description of the Project

"Cekerek Biogas Power Plant" project activity is implemented by "Çekerek Enerji ve Gübre Üretim A.Ş." as per the official documents of the project (e.g. provisional acceptance document, generation license and so on). The project activity is located in Çekerek District, Yozgat Province, Turkey. The project coordinates have been confirmed by the validation team by reviewing EMRA Generation License and KMZ file of the project activity which were provided by the project proponent. The coordinates are as follows:

Latitude	Longitude
40°3'41.04"	35°32'38.70

Also, the coordinates of the farms supplying manure to the project are included in the VCS-PD since the farms are in the project boundary as well. These coordinates have been confirmed by the validation team by evaluating the KMZ file of the local farms which was provided by the project proponent.

As per the provisional acceptance protocol, the start date of the project activity is 29/06/2021. However, since there was sourcing (i.e. manure) problem with the local farmers, the project activity started to receive animal manure on 07/12/2021 and started to reduce GHG emission and generate electricity on that day. This date has been confirmed by the validation team via the official electricity generation records (i.e. EPIAS records) of the project activity. Between 29/06/2021 – 06/12/2021, there was no electricity generation as per these EPIAS records.

The chronology timeline of the project activity is as follows:

Date	Project Milestone

08/04/2021	EIA Decision
10/06/2021	EMRA Generation License
29/06/2021	Provisional Acceptance Protocol
07/12/2021	Start Date of the Electricity Generation (i.e. the start date of the project activity)

The relevant EIA Decision, EMRA Generation License, Provisional Acceptance Protocol, and EPIAS records for assessing the start date of the electricity generation of the project activity have been provided by the Project Proponent.

The project currently receives animal manure from 14 other cattle farms, 5 dairy cattle farms and 4 chicken farms. After animal manure is brought to the project site via the special trucks, the manure is poured to 2 manure receiving tanks and then, to the feeding tank. After these steps, the animal manure is taken to the anaerobic digesters. There are 3 main and parallel anaerobic digesters on the project site. The biogas coming out of there is collected and sent to the gas engine with the installed capacity of 1.501 MWe. By means of this, clean electricity can be generated. This electricity is delivered power mostly to the Turkish grid whilst some is used for auxiliary electricity consumption.

Also, there are solid and liquid fermented product storage units on the project site. The fermented products are given to local people for free. All of these information has been confirmed during the on-site visit dated 24/11/2023.

The key parameters about the technical design of the project are listed below:

Manufacturer and Serial Number of the Weighbridge	ESİT ; Y220080
Type and Capacity of the Anaerobic Digester	EPDM Membrane ; 3,380 m^3 each
Brand and the Power of the Gas Engine	Jenbacher ; 1,501 kWe
Locations and the Serial Numbers of Gas Flow Meters	Before the gas engine: S4071502000 Before the flare unit: VM6-2338012
Brand and the Serial Number of the Gas Analyzer	SWG 100 ; 081429
Brand and the Serial Numbers of the Electricity Meters	Landis Gyr 550 ; Main: 57008965 and Back-up: 57008968
Brand and the Serial Number of the Muffle Furnace	Magmatherm MT-1100-OC-E4 ; E308-211118-01

Brand and the Serial Number of Dry Air Sterilizer	Nükleon NST-55 ; 21ST55339
Brand and the Serial Number of the Scale	OHAUS PR224 ; B110287510

All of these technical details have been confirmed by the validation team via the calibration documents of the monitoring equipment (e.g. flow meters), feasibility study of the installed technology (e.g. anaerobic digesters) and site visit observations.

The first crediting period is between 07/12/2021 – 06/12/2028 (both days included). The estimated annual emission reduction value is 23,909 tCO_2e/year (167,363 tCO_2e in total).

2 VALIDATION PROCESS

2.1 Method and Criteria

Ruby Canyon Environmental has been appointed by “Çekerek Enerji ve Gübre Üretim A.Ş.” to perform the validation of “Cekerek Biogas Power Plant”. Validation was conducted following the Ruby Canyon Environmental procedures created in line with the requirements of the VCS Standard v4.6, CDM Validation and Verification Standard v3.0, and relevant UNFCCC requirements and standard auditing techniques.

A competent assessment team is arranged before the validation process begins. For the purpose of assessing the VCS project activities, the team is chosen to address the technical areas, sectoral scopes and relevant host country experience. To further ensure quality and completeness, a capable of Technical Reviewer is assigned. The validation team visits the site after conducting a desk review, from which a draft report and a list of findings are formed. The evaluation of the results by direct connection with the project proponents is the following step, and the preparation of the validation report is taken in place. This validation report and other supporting documents then undergo an internal quality before submission to VCS.

The validation process included the following independent and objective activities:

- Select a validation team. The validation team was selected according to RCE’s GHG Verification Policies & Procedures to ensure team members are qualified to perform validation activities pertaining to the Project. The validation team consisted of the following individuals:
    - Lead Validator: Öykü YAKUPOĞLU
    - Validator: Mehmet KUMRU
    - Independent Reviewer: Jessica STAVOLE
- Perform a conflict of interest assessment. There were no conflicts of interest identified between RCE and ÇEKEREK ENERJİ VE GÜBRE ÜRETİM A.Ş.
- Submit the Notice of Validation/Verification Services Form to Verra

- Conduct a kick-off meeting with ÇEKEREK ENERJİ VE GÜBRE ÜRETİM A.Ş. to introduce the ÇEKEREK ENERJİ VE GÜBRE ÜRETİM A.Ş. and RCE teams, review the validation objectives, process, VCS requirements, and to confirm the schedule
- Develop a validation plan and sampling plan to be used throughout the validation process. A risk assessment was conducted based on the criteria defined in section 1.2 of this validation report and the evidence provided by ÇEKEREK ENERJİ VE GÜBRE ÜRETİM A.Ş. to demonstrate conformance to VCS Standard and Methodology requirements
- Review the PD against VCS Standard requirements and Methodology requirements. Information in the PD was the primary focus of the validation process. RCE cross-checked information in the PD against supporting evidence and documents to confirm the project start date, location, selection of baseline scenario, demonstration of additionality, ownership, and monitoring plan
- Conduct site visits to selected project activity instances. Details about the selection of project activity instances selected and the subsequent site visits are discussed in more detail in section 2.4 of this validation report below
- Review the accuracy and reasonableness of the estimated emission reductions for the crediting period
- Issue corrective action requests (CARs), additional documentation requests (ADRs), clarification requests (CLs), and forward action requests (FARs) as necessary
- Issue a validation report
- Hold an exit meeting with ÇEKEREK ENERJİ VE GÜBRE ÜRETİM A.Ş.

The key milestones of the validation activity are as follows:

Date	Validation Milestone
08/11/2023	Kick-off Meeting
08/11/2023 – 25/11/2023	Desk Review
24/11/2023	On-site Visit
25/11/2023	Preparation of the Draft Validation Report
05/12/2023	Senior Review
07/12/2023	Preparation of the Final Validation Report

2.2 Document Review

The basis for the validation activity is the VCS-PD version 03 dated 08/11/2023 which was submitted to the validation team on the same date. This VCS-PD report was revised due to the issued CARs and CLs, version 07 dated 10/12/2024 being the final version.

The following actions were involved in the desk review:

- A review of the data and information presented to verify their completeness
- A review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures
- An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions

Document Number	Document Name	Version	Date (DD/MM/YYYY)
D01	Project Description	v03	08/11/2023
D02	Project Description	v04	13/11/2023
D03	Project Description	v05	22/11/2023
D04	Project Description	v06	29/03/2024
D05	Project Description	v07	10/12/2024
D06	ER Calculation Excel Sheet	v03	08/11/2023
D07	ER Calculation Excel Sheet	v04	13/11/2023
D08	ER Calculation Excel Sheet	v05	29/03/2024
D09	ER Calculation Excel Sheet	v06	10/12/2024
D10	Consultancy Agreement between Çekerek Enerji and Climate Solutions	-	10/03/2022
D11	Baseline Surveys with the Farmers	-	2022 - 2023
D12	Emission Factor document published by Ministry of Energy	-	20/09/2022

	and Natural Resources		
D13	Provisional Acceptance Document	-	29/06/2021
D14	KMZ file of the Project Activity and the Farms	-	-
D15	“EIA Not Required” Decision	-	08/04/2021
D16	Technical Documents of the Monitoring Equipment (e.g. flow meters, electricity meters and so on)	-	-
D17	Electricity Generation License	-	10/06/2021
D18	Feasibility Study	-	25/03/2021
D19	First Index Protocols of the Electricity Meters	-	14/07/2021
D20	Calibration Documents of the Flow Meters	-	15/04/2021 23/04/2021
D21	Calibration Document of the Gas Analyzer	-	01/06/2021
D22	Calibration Document of the Weighbridge	-	31/12/2021
D23	Calibration Documents of the Lab Equipment (e.g. muffle furnace, dry air sterilizer, scale, thermocouple and so on)	-	04/06/2021 31/12/2021 11/08/2022

D24	Lab Logbook for SVS data	-	-
D25	Manure Supply Agreements between Çekerek Enerji and the Farmers	-	-
D26	Social Security Records of the Employees	-	-
D27	Declarations about Double Counting and ODA	-	06/11/2023
D28	EPIAS Records	-	06/2021 – 12/2021
D29	Vehicle Licenses	-	-
D30	Signed Waste Water Disposal Statement from Çekerek Municipality	-	06/11/2023
D31	YEKDEM Application	-	29/09/2021
D32	VCS Standard	v4.6	21/03/2024
D33	AMS-III.D Methodology	v21.0	22/09/2017
D34	AMS-I.D Methodology	v18.0	28/11/2014
D35	CDM Tool 03	v03.0	22/09/2017
D36	CDM Tool 05	v03.0	22/09/2017
D37	CDM Tool 06	v04.0	11/03/2022
D38	CDM Tool 07	v07.0	31/08/2018
D39	CDM Tool 08	v03.0	27/11/2015
D40	CDM Tool 14	v02.0	22/09/2017

2.3 Interviews

During the validation period follow-up interviews were realized by the validation team to further analyze the correctness and accurateness of the information provided. The main topics covered during the interview are as follows:

- Installed technology and monitoring equipment (i.e. flow meters, gas analyser, weighbridge, electricity meters and lab equipment) of the project activity
- Project implementation and operation
- Assessment of baseline scenario
- Staff training procedures
- Calibration procedures of the monitoring equipment
- Monitoring system
- Data collection, recording and reporting procedure
- QA/QC procedures
- Eligibility criteria of VCS
- Emission reduction calculations
- Comments from local stakeholders about the implementation of the project activity

The list of people who were interviewed during the physical validation site visit handled on 24/11/2023 is given below:

Reference Number	Means of Interview	Full Name	Title	Organization	Subject
I01	Site Visit	Ömer Oral	Business Manager	Çekerek Enerji	• Installed technology and monitoring equipment of the project activity • Project implementation and operation • Monitoring system

					• Data collection, recording, reporting procedures • QA/QC procedures
I02	Site Visit	Mustafa Bayram Çakır	Laboratory and Site Manager	Çekerek Enerji	• Staff training procedures • Calibration Documents of the monitoring equipment • Lab conditions and lab results
I03	Site Visit	Tayip Yücel	Waste Manager	Çekerek Enerji	• Information of the local farms and transportation of animal manure
I04	Site Visit	İlke Ünal	Operation Manager	Climate Solutions	• Eligibility Criteria • Baseline and Additionality Applicability Conditions, • Emission reductions calculations, • Starting date and crediting period. • Social & Environmental impacts

I05	Site Visit	Ferhat Arslan	Local Stakeholder	Seyit Ahmet Neighbourhood	• Comments about the implementation of the project activity
I06	Site Visit	Yakup Latifhan	Local Stakeholder	Bağlarbaşı Neighbourhood	• Comments about the implementation of the project activity
I07	Site Visit	Kerim Okuroğlu	Local Stakeholder	Bağlarbaşı Neighbourhood	• Comments about the implementation of the project activity
I08	Site Visit	Hasan Çetin	Owner	Hasan Çetin Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation
I09	Site Visit	Mesut Aktaş	Partner	İbrahim Aktaş Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the

					project's implementation
I10	Site Visit	Bekir Zorbilmez	Partner	Yüksel Zorbilmez Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation
I11	Site Visit	Ramazan Elali	Farm Manager	Ali Kırlı Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation
I12	Site Visit	Ahmet Bolat	Owner	Ahmet Bolat Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the

					farm to the project's implementation
I13	Site Visit	Muhammed Elhalit	Farm Manager	Süleyman Bolat Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation
I14	Video Conference	Furkan Çalışkan	Business Manager	Furkan Çalışkan Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation
I15	Video Conference	Kemal Kaya	Business Manager	Kemal Kaya Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline

					scenario at the farm to the project's implementation
I16	Video Conference	Fatih Gövdeli	Owner	Fatih Gövdeli Farm	Agreement with project participant Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation
I17	Video Conference	Selahattin Yıldız	Owner	Selahattin Yıldız Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation
I18	Video Conference	Recep Büyükkaya	Owner	Recep Büyükkaya Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation

I19	Video Conference	Mustafa Yücel	Owner	Mustafa Yücel	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation
I20	Video Conference	Kamil Şahbaz	Owner	Kamil Şahbaz Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation
I21	Video Conference	Servet Yazar	Owner	Servet Yazar Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation

I22	Video Conference	Osman Doğan	Owner	Osman Doğan Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation
I23	Video Conference	Bekir Çekerek	Owner	Bekir Çekerek Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation
I24	Video Conference	Yasin Çapraz	Owner	Yasin Çapraz Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation

I25	Video Conference	Yunus Asıltay	Account Manager	Mehmet Kip Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation
I26	Video Conference	Şevket Kalkan	Owner	Şevket Kalkan Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation
I27	Video Conference	Ahmet Kalkan	Owner	Ahmet Kalkan Farm	• Agreement with project participant • Monitoring procedure, knowledge about project and the baseline scenario at the farm to the project's implementation

2.4 Site Visits

As a part of the validation activities, a physical site visit was performed to the project activity site on 24/11/2023 and details of which can be seen in below:

Points Verified	Source of Information
Checking technical specifications of the Installed Technology and Monitoring Equipment	Physical site visit observations and interview with the business manager
Assessing Baseline Scenario	Physical site visit observations, interview with the local stakeholders and the farmers
Assessing Locations of the Local Farms supplying manure to the project activity	Physical site visit observations and GPS
Assessing monitoring approaches and monitoring data (e.g. SCADA system)	Physical site visit observations, interview with the business manager and the employees
Receiving comments from Local Stakeholders	Interviews with the local stakeholders
Qualification of Animal Farms	Physical farm visit observations, interviews with the local farmers
Receiving comments from Local Farmers	Interviews with the local farmers

All the monitoring parameters furnished by the project proponent have been cross-checked with the electronic records, operational logbooks, and monitoring surveys.

During the on-site visit, the coordinates of the local farms have been confirmed. In this regard, the location of the local farms supplying manure to the project activity are as follows:

Name of the Farm	Type of Livestock	Latitude	Longitude
Ahmet Bolat	Other Cattle	40°4’2.39”	35°31'37.51"
Ali Kırlı	Dairy Cattle	40°3'51.96"	35°28'36.33"
Ali Zorbilmez	Dairy Cattle	40°3'29.79"	35°30'45.50"
Bekir Çekerek	Dairy Cattle	40°4'17.06"	35°29'25.08"
Cemalettin Bozdemir	Other Cattle	40°3'38.47"	35°30'40.03"
İbrahim Aktaş	Dairy Cattle	40°4'10.70"	35°28'58.88"

Osman Doğan	Other Cattle	40°4'1.69"	35°29'1.40"
Recep Büyükkaya	Other Cattle	40°3'49.26"	35°29'3.39"
Selahattin Yıldız	Other Cattle	40°3'56.07"	35°30'50.72"
Süleyman Bolat	Other Cattle	40°3'41.39"	35°31'2.85"
Yüksel Zorbilmez	Other Cattle	40°3'25.82"	35°32'15.98"
Kemal Kaya	Hens	39°42'5.24"	34°30'21.01"
Şevket Kalkan	Other Cattle	39°47'46.89"	34°45'53.69"
Servet Yazar	Other Cattle	39°58'22.45"	35°29'30.83"
Mustafa Yücel	Other Cattle	40°3'33.45"	35°30'27.37"
Abdullah Ersoy	Dairy Cattle	39°43'43.83"	34°35'54.02"
Hasan Çetin	Other Cattle	40°3'59.03"	35°28'29.35"
Ahmet Kalkan	Other Cattle	39°51'28.55"	34°54'24.22"
Fatih Gövdeli	Hens	39°41'10.11"	35°44'25.55"
Kamil Şahbaz	Other Cattle	39°44'41.66"	35°16'47.31"
Furkan Çalışkan	Hens	40°37'53.04"	35° 5'42.09"
Yasin Çapraz	Other Cattle	40°2'41.74"	35°15'57.26"
Mehmet Kip	Hens	40°34'38.37"	34°50'47.64"

After conducting on-site visit at each farm that is part of the project boundary, the VVB confirms that all farms that supply manure are in compliance with the standards of AMS-III.D, v21.0 upon checking the lagoon depth and retention time.

2.5 Resolution of Findings

The validation of this VCS project activity includes the following steps:

- Assessment of the conformity of the actual project activity and its operation with the VCS-PD version 07 and dated 10/12/2024
- A site visit was conducted on 24/11/2023 to assess that all physical features of the project activity proposed VCS PD are in place and that the project participants has operated the project activity as per the VCS PD.

- Assessment of the compliance of the monitoring plan with the monitoring methodology AMS III.D, version 21.0 and AMS.I.D, version 18.0
- Assessment of the compliance of monitoring with the monitoring plan
- Assessment of data and calculation of greenhouse gas emission reductions
- Issuance of the validation report
- Independent technical review
- Approval of the validation report and request of registration
    - In order to ensure transparency a validation protocol, which shows in a transparent manner criteria (requirements), means of validation and VVB findings and final VVB conclusions, is customised for the project activity. All findings and outstanding issues, identified during the desk review and discussed during the site observation, are included in a list of CARs, ADRs, CLs and FARs.
    - During this validation activity, 17 Corrective Action Requests (CARs), 4 Clarification Requests (CLs) and 1 Additional Documentation Request (ADR) were raised all of which were successfully closed. All CARs/ADRs/CLs which are organized in Appendix A, have been resolved by the project proponent via provision of additional supporting evidence and appropriate revision of the VCS PD. Also, a FAR has been issued. For the purpose of completeness, the completed validation protocol is also enclosed to the validation report in Appendix A.

2.5.1 Forward Action Requests

The validation team raises a FAR during validation for actions if the monitoring and reporting require attention and/or adjustment for the next verification period.

According to these principles, a FAR was issued during the validation as in Appendix A.

3 VALIDATION FINDINGS

3.1 Project Details

"Cekerek Biogas Power Plant" project activity is implemented by "Çekerek Enerji ve Gübre Üretim A.Ş." as per the official documents of the project (e.g. provisional acceptance document, generation license and so on). There is an agreement between "Çekerek Enerji ve Gübre Üretim A.Ş." and "CS Climate Solutions Danışmanlık A.Ş." (i.e. the carbon consultancy of the project activity) dated 10/03/2022. As per this agreement, "CS Climate Solutions Danışmanlık A.Ş." is also considered as a project proponent of this project activity.

The signed and sealed letters dated 06/11/2023 about double counting and ODA were received from "Çekerek Enerji ve Gübre A.Ş.". As per these declarations, the project activity has not been registered/seeking registration under other GHG programs. The project portals of CDM, JI, GS, GCC, ICR and BioCarbon were reviewed by the validation team and this project activity is not included on these portals. Also, the project has not been rejected by any other GHG programs as well. The project has not sought or received another form of GHG-related environmental credit, including renewable energy certificates (e.g. I-REC).

Supply Chain (Scope 3) emissions are assessed in line with the VCS Standard 4.6 and Corporate Value Chain (Scope 3) Accounting and Reporting Standard. According to these sources, this section is not applicable since the project does not directly impact emissions associated with any good or service other than the electricity supplied to the national grid. However, according to the statement given at the Corporate Value Chain (Scope 3) Accounting and Reporting Standard "The generator does not account for Scope 3 emissions associated with sold electricity because the emissions are already accounted for in Scope 1. Consequently, the project activity is not related with any Scope 3 emissions.

The eligibility of the project activity has been assessed as per VCS Standard v4.6 and the eligibility criteria has been met for the project activity. In the project activity, manure generated at the neighboring animal farms is collected and treated to capture methane (CH_4), which are among the seven Kyoto Protocol GHGs. The project employs three anaerobic digestion tanks and one gas engine for methane capture and electricity generation and avoids the methane emissions from uncovered anaerobic lagoons present in the baseline scenario. Thus, the project is eligible under this scope. Also, the methodologies applied to the project are AMS-III.D. (Version 21) and AMS-I.D. (Version 18) being approved small scale CDM methodologies, which can be used for programs and projects registering with VCS. Accordingly, the project is eligible.

The project activity is located in Çekerek District, Yozgat Province, Turkey. The project coordinates have been confirmed by the validation team by reviewing EMRA Generation License and KMZ file of the project activity which were provided by the project proponent. The coordinates are as follows:

Latitude	Longitude
40°3'41.04"	35°32'38.70

Before the project activity, animal manure generated at neighbouring animal farms which supplies manure to the biogas plant were left to decay in anaerobic conditions within uncovered anaerobic lagoons for an average retention time period of 2-3 months. This scenario has been confirmed by the validation team by interviewing with the local stakeholders and local farmers during the on-site visit dated 24/11/2023.

The baseline scenario for newly included farms were checked and confirmed as anaerobic decay of animal manure and emission of resulting methane to the atmosphere which complies with the applied Methodology AMS-III.D, version 21.0.

As per AMS-I.D, version 18.0, the baseline scenario is that “The electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid”. The power generated will save the GHG emissions that would have been generated by the fossil fuel based thermal power stations (e.g. coal).

The project is designed to include a single installation only and this was confirmed during the on-site visit dated 24/11/2023.

The project proponent received the necessary official documents to implement this “Çekerek Biogas Power Plant” project (e.g. generation license, EIA and so on). The project has obtained the exemption from the Environmental Impact Assessment (EIA) issued by the Ministry of Environment, Urbanization and Climate Change in Turkey on April 08, 2021, and the Electricity Generation License (EGL) issued by the Energy Market Regulatory Authority (EMRA) in Turkey on June 10, 2021. Hence, the project is in compliance with the laws, statutes, and other regulatory frameworks of the host country, Turkey. Therefore, it can be stated that the project is in compliance with the laws, statutes and other regulatory frameworks of the host country.

As per the provisional acceptance protocol, the start date of the project activity is 29/06/2021. However, since there was sourcing (i.e. manure) problem with the local farmers, the project activity started to receive animal manure on 07/12/2021 and started to reduce methane and generate electricity on that day. This date has been confirmed by the validation team via the official electricity generation records (i.e. EPIAS records) of the project activity. Between 29/06/2021 – 06/12/2021, there was no electricity generation as per these EPIAS records.

The project comes under Sectoral Scope 01: Energy Industries (renewable/non-renewable sources) (Type I Component) and Sectoral Scope 13: Waste Handling and Disposal (Type III Component). Also, the project activity is not a grouped project.

The purpose of the Project Activity is to generate electricity in an efficient, clean, reliable and sustainable way with utmost respect on social and environmental reflections and to reduce emissions by partially substituting the electricity supply of fossil fuel fired power plants in Turkey.

The spatial boundaries of the project activity are clearly described for the project installation and respective emissions reduction through biogas production and electricity generation. The geographical coordinates are also included in the PD.

The project currently receives animal manure from 14 other cattle farms, 5 dairy cattle farms and 4 chicken farms.

During the on-site visit, the coordinates of the local farms have been confirmed. In this regard, the location of the local farms supplying manure to the project activity are as follows:

Name of the Farm	Type of Livestock	Latitude	Longitude
Ahmet Bolat	Other Cattle	40°4’2.39”	35°31'37.51"
Ali Kırlı	Dairy Cattle	40°3'51.96"	35°28'36.33"
Ali Zorbilmez	Dairy Cattle	40°3'29.79"	35°30'45.50"
Bekir Çekerek	Dairy Cattle	40°4'17.06"	35°29'25.08"
Cemalettin Bozdemir	Other Cattle	40°3'38.47"	35°30'40.03"
İbrahim Aktaş	Dairy Cattle	40°4'10.70"	35°28'58.88"
Osman Doğan	Other Cattle	40°4'1.69"	35°29'1.40"
Recep Büyükkaya	Other Cattle	40°3'49.26"	35°29'3.39"
Selahattin Yıldız	Other Cattle	40°3'56.07"	35°30'50.72"
Süleyman Bolat	Other Cattle	40°3'41.39"	35°31'2.85"
Yüksel Zorbilmez	Other Cattle	40°3'25.82"	35°32'15.98"
Kemal Kaya	Hens	39°42'5.24"	34°30'21.01"
Şevket Kalkan	Other Cattle	39°47'46.89"	34°45'53.69"
Servet Yazar	Other Cattle	39°58'22.45"	35°29'30.83"
Mustafa Yücel	Other Cattle	40°3'33.45"	35°30'27.37"
Abdullah Ersoy	Dairy Cattle	39°43'43.83"	34°35'54.02"
Hasan Çetin	Other Cattle	40°3'59.03"	35°28'29.35"
Ahmet Kalkan	Other Cattle	39°51'28.55"	34°54'24.22"
Fatih Gövdeli	Hens	39°41'10.11"	35°44'25.55"
Kamil Şahbaz	Other Cattle	39°44'41.66"	35°16'47.31"
Furkan Çalışkan	Hens	40°37'53.04"	35° 5'42.09"
Yasin Çapraz	Other Cattle	40°2'41.74"	35°15'57.26"
Mehmet Kip	Hens	40°34'38.37"	34°50'47.64"

After animal manure is brought to the project site via the special trucks, the manure is poured to 2 manure receiving tanks and then, to the feeding tank. After these steps, the animal manure is taken to the anaerobic digesters. There are 3 main and parallel anaerobic digesters on the project site. The biogas coming out of there is collected and sent to the gas engine with the installed capacity of 1.501 MWe. By means of this, clean electricity can be generated. Also, there are solid and liquid fermented product storage units on the project site. The fermented products are given to local people for free. All of these information has been confirmed during the on-site visit dated 24/11/2023.

The key parameters about the technical design of the project are listed below:

Manufacturer and Serial Number of the Weighbridge	ESİT ; Y220080
Type and Capacity of the Anaerobic Digester	EPDM Membrane ; 3,380 m^3 each
Brand and the Power of the Gas Engine	Jenbacher ; 1,501 kWe
Locations and the Serial Numbers of Gas Flow Meters	Before the gas engine: S4071502000 Before the flare unit: VM6-2338012
Brand and the Serial Number of the Gas Analyzer	SWG 100 ; 081429
Brand and the Serial Numbers of the Electricity Meters	Landis Gyr 550 ; Main: 57008965 and Back-up: 57008968
Brand and the Serial Number of the Muffle Furnace	Magmatherm MT-1100-OC-E4 ; E308-211118-01
Brand and the Serial Number of Dry Air Sterilizer	Nükleon NST-55 ; 21ST55339
Brand and the Serial Number of the Scale	OHAUS PR224 ; B110287510

All of these technical details have been confirmed by the validation team via the calibration documents of the monitoring equipment (e.g. flow meters), feasibility study of the installed technology (e.g. anaerobic digesters) and site visit observations.

The project activity contributes 4 SDGs:

- SDG 7 Affordable & Clean Energy: The project is expected to export electricity generated by the gas engine (SDG 7.2.1), 11,258 MWh clean electricity/year (as per the EMRA Generation License)

- SDG 8 Decent Work & Economic Growth: Around 20 employees created (SDG 8.0) (currently 16 employees) (as per the social security records of the employees)
- SDG 13 Climate Action: 23,909 tCO2e estimated emission reductions/year (SDG 13.0) (as per the relevant calculations of AMS-III.D, AMS-I.D and relevant tools)
- SDG 17 Partnerships for the Goals: YEKDEM Application (The relevant evidence document has been provided by the project proponent. Please refer to Section 2.2 of this validation report.) (SDG 17.17.1).

The project is twice renewable. Hence, the total crediting period is 21 years. The first crediting period is between 07/12/2021 – 06/12/2028 (both days included). The estimated annual emission reduction value is 23,909 tCO2e/year (167,363 tCO2e in total). Since the estimated annual emission reduction value is below 300,000 tCO2e, the scale of the project activity is "project".

In summary, the description of the project activity in the VCS PD is accurate, complete, and provides an understanding of the nature of the project.

3.2 Safeguards

3.2.1 No Net Harm

The project activity is not on forest land. There are agricultural lands around the project area; however, since the project is constructed on water and, no reservoir, no negative impacts are expected on agricultural land. The operation of equipment complies with the Turkish Standards.

Regarding the wastes, hazardous wastes are handled appropriately in closed containers and transported by licensed transporters to the licensed processing and disposal facilities. Wastewater is collected through within the septic tank and is transferred through these wage truck (The relevant evidence document has been provided from the Çekerek Municipality). Domestic wastes are transferred by the provincial administration. There is not any negative environmental or socio-economic impact.

The EIA Not Required Decision dated as 08/04/2021 by the Yozgat Governorship Provincial Directorate of Environment and Urbanization was also provided by the Project Proponent. A "no-EIA required" decision is only given to the project that cause no harm on the environment by the Ministry of Environment, Urbanization and Climate Change in Turkey. Besides that, the waste storage areas (hazardous, waste water and domestic) have been seen during the on-site visit dated 24/11/2023.

3.2.2 Local Stakeholder Consultation

The Local Stakeholder Consultation has been conducted by the Project Proponents on 20/07/2022 via Microsoft Teams. 33 local stakeholders joined to the meeting. The comments of them were included in Section 2.2 of the VCS PD. There were no negative comments received from the local stakeholders.

There had not been any complaint raised by the interviewed local stakeholders during the physical validation site visit as detailed in Section 2.3. The local stakeholders as stated in Section 2.3 above were interviewed about the following issues and there had not been any complaint by the interviewed local stakeholders during the on-site visit:

- Dust emissions during construction and operation periods
- Impacts on the local farms (related to animal manure)
- Impacts of the fermented products
- Sufficiency of local employment (The interviewed local stakeholders were pleased about the provided local employment opportunities by the Project Proponent)
- Waste management practices implemented by Project Proponent

It was also concluded that the grievance mechanism (via telephones and face-to-face communication) is in place and this was also confirmed by the interviewed local stakeholders during the on-site visit.

3.2.3 Environmental Impact

The project activity is covered by a Preliminary Environmental Impact Analysis (PIF). A full EIA is not required since the scale of the project is below the applicable. The PIF has not identified any significant environmental impacts. Also, "EIA Not Required" decision dated 08/04/2021 has been taken for the project activity.

Besides that, the waste storage areas (hazardous, waste water and domestic) have been seen during the on-site visit dated 24/11/2023.

3.2.4 Public Comments

No comments were received during the public comment period between 29/06/2023 – 29/07/2023[1].

3.2.5 AFOLU-Specific Safeguards

N/A (The project activity is not an AFOLU project.)

3.3 Application of Methodology

3.3.1 Title and Reference

"AMS-III.D: Methane recovery in animal manure management systems, version 21.0"[2] and "AMS-I.D: Grid-connected renewable electricity generation, version 18.0"[3] have been applied to the project activity. Since the project activity includes both waste handling part (Type III component) and electricity generation part (Type I component), the selected methodologies have been confirmed by the validation team.

[1] https://registry.verra.org/app/projectDetail/VCS/4403

[2] https://cdm.unfccc.int/methodologies/DB/H9DVSB24O7GEZQYLYNWUX23YS6G4RC

[3] https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQQOFQQH4SBK

Moreover, the latest versions of the methodologies have been selected by the Project Proponent at the time of the validation.

The selected tools and their latest versions related to above methodologies are as follows:

- Tool 03: Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion, version 03.0[4]
- Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0[5]
- Tool 06: Project emissions from flaring, version 04.0[6]
- Tool 07: Tool to calculate the emission factor for an electricity system, version 07.0[7]
- Tool 08: Tool to determine the mass flow of a greenhouse gas in a gaseous stream, version 03.0[8]
- Tool 14: Project and leakage emissions from anaerobic digesters, version 02.0[9]

Ruby Canyon Environmental confirms that the selected methodologies are applicable to the project activity, and the versions will be valid at the time of submitted of the project activity for the validation.

3.3.2 Applicability

Ruby Canyon Environmental has assessed the relevant information contained in the PD, on-site audit and evidence obtained against the application criteria listed in the methodologies, AMS-III.D and AMS-I.D. The applicability conditions of the methodologies are justified as below:

For AMS-III.D, version 21.0:

- The livestock population in the farms is managed under confined conditions. As per “Regulation on Livestock Farms”, this condition shall be followed. Also, during the on-site visit dated 24/11/2023, the confined conditions were confirmed by the validation team.

[4] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-03-v2.pdf/history_view

[5] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v1.pdf/history_view

[6] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-06-v1.pdf/history_view

[7] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v2.pdf/history_view

[8] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-08-v3.0.pdf/history_view

[9] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-14-v1.pdf/history_view

- Manure obtained after treatment are not discharged into natural water resources. This condition was confirmed during the on-site visit dated 24/11/2023 with farm visit observations and interviews with the local farmers.

- The annual average temperature of Yozgat province is 9.7 ºC[10][11].
- In the baseline scenario, the retention time of manure waste in the anaerobic treatment system is greater than 1 month, and the depths of anaerobic lagoons are at least 1 m. This condition was confirmed during the on-site visit dated 24/11/2023 with farm visit observations and interviews with the local farmers. Also, it has been confirmed that no methane recovery and destruction takes place in the baseline scenario.
- The residual waste from the animal manure management system of the project activity is sent out and used for agriculture utilization aerobically in the nearby farmlands. The residual waste from the animal manure management system of the project activity is first stored in fermented product storage units, then sent out and used for agriculture utilization aerobically in the nearby farmlands. There are two (one liquid, one solid) fermented product storage units installed at the project site, allowing the products to be stored in aerobic conditions before being taken for application to agricultural plots. This condition was confirmed during the on-site visit dated 24/11/2023 with interviews with the local stakeholders.

- The project activity utilizes biogas for electricity generation. A flaring system is also installed to the flare the gas in case of emergencies. This condition was confirmed during the on-site visit dated 24/11/2023 with site visit observations.

- The manure after removal from the animal barns is transferred into the anaerobic digesters via waste receiving units within same day. This situation has been confirmed with interviews with the employees.

- In the project activity, methane is not recovered from landfills. Also, the project does not involve wastewater treatment. Moreover, there is no composting of animal manure procedure. Finally, the project does not involve co-digestion of animal manure and other organic matter. All of these situations have been confirmed via the site visit observations.

- The recovered biogas is used mainly to produce electricity. This situation has been confirmed via the results in the SCADA system during the on-site visit.

- The project activity is a greenfield project. This situation has been confirmed via the interviews with the local stakeholders.

[10] https://www.mgm.gov.tr/veridegerlendirme/il-ve-ilceler-istatistik.aspx?m=YOZGAT

[11] The PP has gathered official mean annual temperature data from the nearest meteorological centers to the baseline animal manure management systems, as provided by the Turkish State Meteorological Service Office. The detailed data have been submitted to the VVB and it indicates the annual average temperature at each baseline site exceeds 5 ºC. For simplicity, the annual average temperature for YOZGAT, the province, where the baseline sites are located, is included here.

- The estimated annual emission reduction is 23,909 tCO_2e/year. Since it is lower than 60,000 tCO_2e, AMS-III.D can be used for the project activity.

For AMS-I.D, version 18.0:

- The project activity is a greenfield project. This situation has been confirmed via the interviews with the local stakeholders.
- The project activity is a biogas-to-energy.
- There are no non-renewable components in the project activity. This situation has been confirmed via the site visit observations.
- The project does not involve co-digestion of animal manure and other organic matter. This situation has been confirmed via the site visit observations.
- Biomass is not used for the project activity.

Moreover, the applied tools have been assessed with on-site visit and evidence obtained against the application criteria listed in the referred tools which are Tool 03, Tool 05, Tool 06, Tool 07, Tool 08, and Tool 14:

- Tool 03 Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion (version 03.0): CO_2 emissions from fossil fuel are calculated based on the quantity of fuel combusted by the vehicles for transporting manure.
- Tool 05 Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0): The proposed project activity may only purchase electricity from the Turkish grid and no captive power plant is installed at the site of electricity consumption (Scenario A). Also, electricity generated is supplied to the grid (Scenario I). Moreover, only CO_2 emissions are accounted for the calculations following this tool.
- Tool 06 Project emissions from flaring (version 04.0): Methane is the highest concentration in the produced biogas. This situation can be confirmed via the results coming from the gas analyzer. Also, there is no use of auxiliary fuels and the residual gas will be flared and/or if the gas engines are not functioning.
- Tool 07 Tool to calculate the emission factor for an electricity system (version 07.0): Because the project activity generates electricity to the national grid according to the connection agreement, this tool can be applied to calculate the emission factor.
- Tool 08 Tool to determine the mass flow of a greenhouse gas in a gaseous system (version 03.0): Mass flow will be determined for CH_4. The time frame will be minute basis. If flare unit is used, Tool 08 will be used to calculate the project emissions related to the flare.
- Tool 14 Project and leakage emissions from anaerobic digesters (version 02.0): $EF_{CH4,default}$ value can be taken from this tool.

Therefore, Ruby Canyon Environmental confirms that the selected methodologies and the tools are applicable to the project activity, and the versions will be valid at the time of submitted of the project activity for the validation.

3.3.3 Project Boundary

All the units of the project activity are in line with the requirements of the methodologies applied. Figure 24 in the VCS-PD indicates clearly the project boundary and the relevant features.

As per the applied methodologies of AMS-III.D (version 21.0) and AMS-I.D (version 18.0), the spatial extent of the project boundary encompasses:

- The livestock
- Animal manure management systems (including the centralized manure treatment plant)
- Facility which recovers and utilizes methane
- The project power plant and all power plants connected physically to the electricity system that the project power plant is connected to

Also, GHG sources related to the project activity are as follows:

- Direct emissions from the manure treatment processes (Baseline)
- Emissions from electricity consumption/generation (Baseline)
- Physical leakage of biogas (Project)
- Emissions from on-site electricity use or fossil fuel use (Project)
- Emissions from flaring of the gas stream (Project)
- Emissions from incremental transportation distances (Project)

GHG sources have been confirmed by the validation team. Since animal manure is taken from nearby farms, the direct emissions from the manure treatment processes is a GHG source for the baseline scenario. Also, the quantity of electricity produced using biogas technologies would have been produced by power plants connected to the national grid in Turkey which is dependent on fossil fuel resources. So, emissions from electricity consumption/generation is another GHG source for the baseline scenario. The project emissions stated AMS-III.D (version 21.0) are included in the GHG calculations except “Emissions from storage of manure” since the manure after removal from the animal barns will be transferred into the anaerobic digesters via pre-treatment pool within same day, so the storage time of manure is less than 24 hours. This information was confirmed with interviewing with the employees during the on-site visit.

The validation team confirmed that there are no emission sources that are not addressed by the applied methodologies which are expected to contribute more than 1% of the annual emission reductions.

3.3.4 Baseline Scenario

As per the methodology of AMS-III.D, version 21.0, the baseline scenario is that "In the absence of the project activity, animal manure is left to decay anaerobically within the project boundary and methane is emitted to the atmosphere".

In the VCS-PD, the four step approach is taken due to the requirements of General guidelines for SSC CDM methodologies. In summary, all alternative baseline scenarios (as per The table 10.14 in Chapter 10, Volume 4 of 2019 Refinement of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories) are compared with each other. For the manure management systems, only "uncovered anaerobic lagoon" and "anaerobic digesters" are permitted by Turkish laws and regulations. The anaerobic digesters undertaken without being registered as a CDM/VCS project activity is eliminated due to technological (e.g. complete nature of anaerobic digestion technology) and investment barriers (e.g. capital outlay for the installation and operation of anaerobic digester systems).

Only one alternative scenario is left. Therefore, the project activity can be considered as eligible under AMS-III.D, version 21.0 with the baseline scenario being uncovered anaerobic lagoons as manure management system.

As per AMS-I.D, the baseline scenario is that "The electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid". The power generated will save the GHG emissions that would have been generated by the fossil fuel based thermal power stations (e.g. coal).

In summary, the indicated baseline scenario for the electricity generation in the PD is found appropriate by the validation team.

3.3.5 Additionality

As per AMS-III.D: "Project activities may demonstrate the additionality by showing that there is no regulation in the host country, applicable to the project site, that requires the collection and destruction of methane from livestock manure. If so, it is not required to apply the 'Guidelines on the demonstration of additionality of small-scale project activities".

There is no legal regulation in Turkey that requires the collection and destruction of methane from livestock manure. Therefore, the above situation can be considered for the proposed project activity.

Moreover, again as per AMS-III.D: "This additionality condition also applies to Greenfield project activities. Furthermore, for project activities applying this methodology in combination with a Type I methodology, that has an energy component whose installed capacity is less than 5 MW, this procedure for additionality demonstration also applies to that component".

There is only one gas engine on the project site and its installed capacity is 1.501 MWe (< 5MW). Therefore, the above situation can be considered for the proposed project activity as well.

In summary, as per the applied methodology AMS-III.D, version 21.0, the project activity is deemed additional.

3.3.6 Quantification of GHG Emission Reductions and Removals

The emission reduction calculation estimations have been done in the PD as per the latest approved version of the methodologies AMS-III.D (version 21.0) and AMS-I.D (version 18.0).

The operating margin emission factor and build margin emission factor have been taken from the most recent publicly available source for Turkey published on Ministry of Energy and Natural Resources website[12]. As per this document:

$EF_{grid,OM,y}$ = 0.7424 tCO_2/MWh

$EF_{grid,BM,y}$ = 0.3680 tCO_2/MWh

Therefore, the combined margin emission factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

$$EF_{grid,CM,y} = (0.7424) \times (0.50) + (0.3680) \times (0.50) = 0.5552\ tCO_2/MWh$$

Baseline emissions are calculated using the proper equations in the applied methodologies AMS-III.D (version 21.0) and AMS-I.D (version 18.0):

- Since direct measurement of quantity of manure and its specific volatile solids contents can be determined by the Project Proponent, Equation (5) of AMS-III.D (version 21.0) is used to calculate the baseline emissions related to Type III component of the project activity.

- For Equation (5), the used values have been confirmed by the validation team via the relevant evidence documents:

Parameter	Value	Reference Document
GWP_{CH4}	28	IPCC Fifth Assessment Report
D_{CH4}	0.00067 t/m^3	AMS-III.D, version 21.0
$Q_{manure,LT,y}$	Please see “Validation ER” Excel sheet	Data from Calibrated Weighbridge
$SVS_{j,LT,y}$	Please see “Validation ER” Excel sheet	Data from Calibrated Lab Equipment

[12] https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUESEmisyonFktr/Belgeler/Bform2020.pdf

MCF_j	0.67	2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories (IPCC 2019) (in accordance with Information Required to Determine Climate Zones According to Chapter 3 of Volume 5 Current Guideline of IPCC 2019, page 10.129, Yozgat is considered as Cool Temperate Dry as its climate zone)
B_0 for dairy cattle, other cattle and hens	Dairy: 0.24 m^3 CH_4/kg-dm Other: 0.18 m^3 CH_4/kg-dm Hens: 0.39 m^3 CH_4/kg-dm	2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories (IPCC 2019) - For Middle Region – High Productivity System
UF_b	0.94	AMS-III.D, version 21.0

- As conclusion, baseline emissions from AWMS is calculated as 19,652 tCO_2e/year.
- As per AMS-I.D, the baseline emission related to the electricity generation is calculated using Equation (1). The relevant values have been confirmed by the validation team via the relevant evidence documents:

Parameter	Value	Reference Document
$EG_{Pj,y}$	11,258 MWh	Generation License
$EF_{grid,CM,y}$	0.5552 tCO_2/MWh	Ministry of Energy and Natural Resources and Tool 07

- As conclusion, baseline emissions from the electricity generation is calculated as 6,250 tCO_2e/year.
- Therefore, the estimated annual baseline emission value is (19,652 + 6,250) = 25,902 tCO2e/year.

Project emissions are calculated using the proper equations in the applied methodologies AMS-III.D (version 21.0) and AMS-I.D (version 18.0):

- Equation (6) of AMS-III.D (version 21.0) is used to calculate the project emissions related to the proposed project activity.

- To calculate emissions due to physical leakage of biogas, Equation (2) and Equation (3) of Tool 14 are used. The relevant values are as follows:

Parameter	Value	Reference Document
$Q_{biogas,y}$	6,214,140 Nm^3	SCADA system
Q_{CH4}	2,498 tCH_4	SCADA system
EF_{CH4} (Digesters with steel or lined concrete or fiberglass digesters and a gas holding system (egg shaped digesters) and monolithic construction)	0.028	Tool 14

- As conclusion, project emissions from the physical leakage of biogas is calculated as 1,959 tCO_2e/year.

- To calculate emissions from the use of fossil fuel or electricity for the operation of the installed facilities, Equation (1) of Tool 05 is used. Since no electricity is purchased from the grid so far, the relevant project emissions are considered as zero at the time of validation.

- To calculate emissions from flaring, Equation (15) of Tool 06 is used. Since flare is not used so far, the relevant project emissions are considered as zero at the time of validation.
- To calculate fossil fuel used from emergency diesel generator, Equation (1) and Equation (4) of Tool 03 is used. Since the emergency diesel generator is not used so far, the relevant project emissions are considered as zero at the time of validation.

- To calculate emissions from incremental transportation, Equation (3) of AMS-III.AO (version 01.0) is used. The relevant values are as follows:

Parameter	Value	Reference Document
Q_y	59,725 tonnes	Calibrated Weighbridge Data
CT_y	18 tonnes/truck	Project Proponent
DAF_w	41 km/truck	KMZ file of the farms
$EF_{CO2,transport}$	0.245 $kgCO_2$/km	Tool 12

- As conclusion, project emissions from transportation is calculated as 34 tCO_2e/year.
- Therefore, the estimated annual project emission value is (1,959 + 34) = 1,993 tCO2e/year.

No leakage emissions are taken into account for the project activity. The solid digestate is not disposed of in a SWDS or a stockpile; instead, it is distributed directly to local farmers for agricultural us efor free. Therefore, the calculation of LEstorage,y for the solid digestate is not applicable.

In summary, the estimated annual emission reduction value is calculated as follows:

$$ER_y = BE_y - PE_y - LE_y$$

$$ER_y = 25{,}902 - 1{,}993 - 0$$

$$ER_y = 23{,}909 \ tCO_2e/year$$

The calculations and the selection of the values are accepted by the validation team.

3.3.7 Methodology Deviations

N/A (No methodology deviation is applied.)

3.3.8 Monitoring Plan

The monitoring parameters are in line with the applied methodologies and include the following:

- $Q_{manure,LT,y}$: Quantity of manure treated from livestock type LT at animal manure management system j (tonnes-dm/year)
- Q_y: Quantity of raw manure co digested in the year y (tonnes-wet basis/year)
- $SVS_{LT,y}$: Specific volatile solids content of animal manure from livestock type LT and animal manure management system j in year y (tonnes VS/tonnes-dm)
- $B_{0,LT}$: Maximum methane producing potential of the volatile solid generated for animal type 'LT' (m^3CH_4/kg-dm)
- CT_y: Average truck capacity for transportation (tonnes/truck)
- DAF_w: Average incremental distance for raw solid manure transportation (km/truck)
- $EG_{PJ,y}$: Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/year)
- $EC_{PJ,J,y}$: Quantity of electricity consumed by the project electricity consumption source the national grid in year y (MWh/year)
- $EF_{grid,y}$: Emission factor for the Turkish National Grid (tCO_2e/MWh)

- $TDL_{j,y}$: Average technical transmission and distribution losses for providing electricity to source j in year y (fraction)

- $Q_{biogas,y}$: Amount of biogas collected at the digester outlet in year y (Nm^3)

- $Q_{CH4,y}$: Quantity of methane produced in the anaerobic digester in year y (tCH_4/y)

- MCF_j: Annual methane conversion factor (MCF) for the baseline animal manure management system j (fraction)

- DM: Residue remaining after water is removed from waste material by evaporation; dry matter (%)

- η_{flare}: Flare efficiency in the year y (%)

- $V_{t,db}$: Volumetric flow of the gaseous stream in time interval t on a dry basis (m^3 dry gas/h)

- $V_{CH4,t,db}$: Volumetric fraction of methane in a time interval t on a dry basis (fraction)

- $Flame_m$: Flame detection of flare in the minute m (on or off)

- T_t: Temperature of the gaseous stream in time interval t (K)

- P_t: Pressure of the gaseous stream in time interval t (Pa)

- EG_y: Total electricity generated from the recovered biogas in year y (MWh)

- EE_y: Energy conversion efficiency of the project equipment (%)

The applied methodology and tools refer these monitoring parameters. Ruby Canyon Environmental has checked Data Unit, Description, Source of Data, Description of the Measurement Method, Frequency of Monitoring, Value Applied, Monitoring Equipment, QA/QC Procedures, and Calculation Method of these parameters in the applied methodology and tools. All information for the monitoring parameters has been indicated correctly in the VCS-PD.

Measuring data for electricity and methane volumes is done with calibrated meters. According to the monitoring methodology, the accumulated data on electricity meters, flow meters, thermometers, and gas analyzer is recorded. All the data will be archived electronically and kept at least two years after the last crediting period.

The technical details of the monitoring equipment are included in Section 1.11 of the VCS-PD and Section 3.1 of the validation report. All technical details have been confirmed by the validation team with reviewing the calibration documents of the equipment. Also, all meter readings and equipment status are collected automatically.

Ruby Canyon Environmental can certify that the list of parameters to be monitored is complete and consistent with AMS-III.D (version 21.0) and AMS-I.D (version 18.0), and that the monitoring plan adheres to the monitoring methodology used.

The validation team confirms that the monitoring plan can be properly implemented, that all monitoring arrangements are feasible within the project design as per the inspections of the on-site visit, and that the means of implementation of the monitoring plan, including data management and quality assurance and quality control procedures, are sufficient to ensure that the ERs to be achieved by the project activity can be properly reported and verified through document review and interview with the project owner.

3.4 Non-Permanence Risk Analysis

N/A (The project activity is not an AFOLU project.)

4 VALIDATION OPINION

Ruby Canyon Environmental has performed the validation activity of “Cekerek Biogas Power Plant” which is a proposed VCS project with the reference number “4403” for the crediting period between 07 December 2021 – 06 December 2042 (21 years, twice renewable). The first crediting period of the project activity is between 07 December 2021 – 06 December 2028 (both days included). The scope of the activities covers the validation and certification of GHG emissions reductions reported in the VCS Project Description Report Version 07 dated 10 December 2024.

CS Climate Solutions Danışmanlık A.Ş. is responsible for the preparation of the GHG emissions data and the reported GHG emissions reductions of the project on the basis set out within the project Monitoring Plan indicated in the final VCS PD. The development and maintenance of records and reporting procedures in accordance with that plan, including the calculation and determination of GHG emission reductions from the project, is the responsibility of the management of the Project. The development and maintenance of the records and the related monitoring procedures are in accordance with the Project Description Report Version 07.

Ruby Canyon Environmental concludes that the GHG assertion is free of material misstatement. The emission reductions calculated can be considered in conformance with the:

- The Article 12 of the Kyoto Protocol
- VCS Standard, v4.6, Issued 19 September 2019; updated 21 March 2024 (VCS Standard)
- VCS Program Guide, v4.4, Issued 19 September 2019; updated 29 August 2023
- VCS Validation and Verification Manual, v3.2, 19 October 2016
- CDM Validation and Verification Standard for Project Activities, v3.3
- AMS-III.D: Methane recovery in animal manure management systems, v21.0
- AMS-I.D: Grid connected renewable electricity generation, v18.0

- All associated methodological tools
- Environmental and social issues related to the Sectoral Scope 01: Energy industries (renewable/ non-renewable sources) and Sectoral Scope 13: Waste handling and disposal
- ISO 14064-3:2019 Greenhouse Gases – Part 3: Specification with guidance for the verification validation of greenhouse gas statements

The validation has been performed by a validation team consisting of “Öykü YAKUPOĞLU as the Lead Validator, Mehmet KUMRU as the Validator and Jessica STAVOLE as the Independent Reviewer” and the project activity was checked against the applicable rules and regulations of VCS version 4.6.

The validation activity can confirm that:

- the project is implemented and operated as per the VCS PD (version 07 dated 10/12/2024)
- the monitoring complies with the monitoring plan in the VCS PD
- the VCS-PD and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS and CDM requirements
- the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately
- the monitoring system is in place and generates GHG emission reductions data
- the GHG emission reductions are calculated without material misstatements

Ruby Canyon Environmental hereby confirms that the project activity “Cekerek Biogas Power Plant” in Turkey, is implemented in accordance with the PD version 07 and dated 10 December 2024. The monitoring system is in place and the emission reductions are estimated without material misstatements as per the applied approved methodologies, which are “AMS-III.D, version 21.0” and “AMS-I.D, version 18.0”.

Validated GHG emission reductions and removals in the above period:

Year	Estimated GHG emission reductions or removals (tCO_2e)
2021 (07/12/2021 – 31/12/2021)	1,637
2022	23,909
2023	23,909
2024	23,909
2025	23,909
2026	23,909

2027	23,909
2028 (01/01/2028 – 06/12/2028)	22,272
Total estimated ERs	167,363
Total number of crediting years	7
Average annual ERs	23,909

APPENDIX A: VALIDATION PROTOCOL

Corrective Action Request (CAR), Non-Material Finding (NMF), Additional Documentation Request (ADR), or Clarification Request (CL) #	Finding and Date (First findings dated 09/11/2023)	Section of Protocol/ Methodology or Program Document	Project Developer Response and Date	RCE response and Date (Second findings dated 18/11/2023)	Additional Project Developer Response and Date	Additional RCE Response and Date	Open or Closed
CAR 1	In Section 1.1: a) As per the provisional acceptance protocol, there are 2 gas engines at the project site. However, in Section 1.1, there are multiple parts where 1 gas engine is stated. Please correct the contradiction. Also, the same issue is present in some other sections in the VCS PD (e.g. Section 1.3, Section 1.11, in Figure 1 and so on). b) The start date of the project activity is 29/06/2021 as per the provisional acceptance protocol. However, the start date is accepted as 07/12/2021 in the	Section 1.1	a) This project has a licenced capacity of 3.002 MWe according to the electricity generation licence. However, only one of the gas engines with an electricity generation capacity of 1.501 MWe is installed at the project site as per the provisional acceptance protocol. Thus, the installed capacity of the project is reported as 1.501 MWe and number of installed gas engines is reported as 1. b) Reasoning behind the determination of start date has been included to the Section 1.1. c) Information in Table in Section 1.1 has been corrected.	a) Ok Closed (The clarification was made.) b) Ok Closed (The relevant information was included in Section 1.1.) c) Ok Closed (Table 1 was revised accordingly.)			Closed

	report. Please indicate the reason briefly in Section 1.1. c) Please correct the information indicated in the table in Section 1.1.						
CAR 2	Please correct the "Title" of Dr. Umut Önder in Section 1.5.	Section 1.5	Relevant section has been revised.	Ok Closed (The title was corrected.)			Closed
CAR 3	In Section 1.7, Appendix 1 and Appendix 2 are referred for the EIA and generation license of the project. However, these ones are not the right sections. Please correct the relevant references in Section 1.7 (Section 1.11 has the same issue.).	Section 1.7	Contradicting references has been corrected.	Ok Closed (The relevant references were corrected.)			Closed
CAR 4	In Section 1.10: a) Please apply the "round down function" for the baseline emissions and emission reductions throughout the PD and in the Excel sheet (e.g. (27,370/365)*25= 1,874.657… shall be 1,874.). b) Please	Section 1.10	a) "rounddown" function for the baseline and ER calculations and "roundup" function for the project emission calculations have been applied through the ER sheet and PD. b) End date of the first creditin period has been revised in Section 1.10.	a) Ok Closed (Section 1.10 was revised accordingly.) b) Ok Closed (The end date was corrected in Section 1.10.)			Closed

	correct the end date of the first crediting period in Section 1.10.						
CAR 5	In Section 1.11: a) In Appendix 3, there is no evidence for the date "07/12/2021". Please remove the reference from Section 1.11. b) The number of gas engines and the installed capacity of each of them are indicated wrongly in Section 1.11. Please correct the contradictions. c) Please correct the value of "28,178 tonnes manure/day" in Section 1.11. d) On page 13, one animal manure receiving tank is stated. However, in Figure 1, there are 2 manure receiving tanks. Please correct the contradiction. e) Please include technical details of the gas analyzer (if it is combined with the flow meters, please include	Section 1.11	a) Irrelevant reference has been removed. b) Number of gas einges has been reported in line with the provisional acceptance document. c)The value has been corrected as 77 tons manure/day. d) Number of manure receiving tanks has been corrected as two. e) Technical details of the gas analyzer has been included to the Section 1.11.	a) Ok Closed (The irrelevant reference was removed.) b) Ok Closed (The clarification was made.) c) Ok Closed (The relevant value was corrected in Section 1.11.) d) Ok Closed (Section 1.11 was revised accordingly.) e) Ok Closed (The technical details of the gas analyzer were included in Section 1.11.)			Closed

	this information) in Section 1.11.						
CAR 6	The livestock types of some farms are indicated differently in Section 1.12 compared to the ER Excel sheet. Please correct the contradictions.	Section 1.12	Type of Livestock column has been revised to eliminate contradictions in Section 1.12.	Ok Closed (The relevant types were corrected in Section 1.12.)			Closed
CAR 7	Please use the VCS template for Section 1.16.	Section 1.16	Section 1.16 is reported as per the VCS Project Description Template v4.2	Ok Closed (The clarification was made.)			Closed
CAR 8	In Section 1.17: a) In Section 1.17, "...five of the UN Sustainable Development Goals (SDGs)..." is stated. However, 4 SDGs are demonstrated. Please correct the contradiction. b) Please correct the contribution of SDG 7 in Section 1.17.	Section 1.17	a) The relevant statement has been corrected. b) contribution of SDG 7 has been reported as per the installed capacity of the project.	a) Ok Closed (The relevant sentence was corrected in Section 1.17.) b) Please indicate the contribution as "estimated electricity generation in a year".	b) Estimated electricity generation has been indicated as contribution to SDG 7.	b) Ok Closed (The relevant value was revised.)	Closed

CAR 9	In Section 2.2: a) Please revise "Çekerek Energy Facility" statements to "Cekerek Biogas Power Plant" in Section 2.2. b) Please revise "Image" statements to "Figure" in Section 2.2 and continue the numbering from Section 2.1.	Section 2.2	a) The relevant statements have been revised, as requested. b) Numbering and name of figures has been revised throughout the document.	a) Ok Closed (The relevant statements were corrected in Section 2.2.) b) Ok Closed (The relevant statements were corrected in Section 2.2.)			Closed
CAR 10	Please complete Section 2.4.	Section 2.4	Section 2.4 has been revised to include public comment details.	Ok Closed (Section 2.4 was completed.)			Closed
CAR 11	In Section 3.2: a) Please include "cattle farm owners" in " Since there is no legal regulation to mandate the poultry farm owners..." sentence on page 56 (Please also include this into the justification for applicability condition 4c. Also, in Section 3.4.). b) Please include a justification for "20% dry matter content" in "Paragraph 4-c" condition on page 57. c) Please correct the installed capacity of the project activity throughout the PD. d) Please re-consider the	Section 3.2	a) The justifications have been revised to include cattle farms as well as chicken farms. b) Because this project obeys by the maximum of 45 condition, providing a justification for "20% dry matter content" is not necessary. c) Installed capacity of the plant is reported in line with the provisional acceptance document. d) Justifications for "Paragraph 8 and 9" conditions of AMS-I.D in Section 3.2 has been revised as requested. e) Tool number has been corrected as Tool05.	a) Ok Closed (The relevant statements were corrected.) b) Ok Closed (The clarification was made.) c) Ok Closed (The clarification was made.) d) Ok Closed (The relevant statements were revised.) e) Ok Closed (The number of the tool was corrected.)			Closed

	justifications for "Paragraph 8 and 9" conditions of AMS-I.D in Section 3.2. e) Please correct the tool number for "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" on page 64.						
CAR 12	In Section 3.3: a)Please revise the spatial extent with considering AMS-III.D and AMS-I.D. b) Please consider the project boundary of AMS-I.D methodology as well in Figure 24.	Section 3.3	a) Spatial extent of project boundry has been revised as defined in the applied Methodologies AMS-III.D and AMS-I.D. b) Figure related to the project boundry has been revised to reflect boundries of AMS-III.D and AMS-I.D.	a) As per AMS-I.D "The spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system". Therefore, please add "all power plants" statement as well for the spatial extent of the project activity. b) Please see	a) Spatial extent of project related to AMS-I.D methodology has been revised as "The project power plant and all power plants connected physically to the electricity system that the project power plant is connected to". b) Related figure (Fig 22) has been revised to include the "grid power	a) Ok Closed (Section 3.3 was revised accordingly.) b) Ok Closed (The figure was closed accordingly.)	Closed

				option (a) of this finding.	plants" since these power plants are connected to the electricity system the project power plant is connected to (grid).		
CAR 13	The total installed capacity of the project activity is stated as 4 MW in Section 3.5. Please correct the value.	Section 3.5	Installed capacity of the project has been corrected in Section 3.5.	Ok Closed (The installed capacity was corrected in Section 3.5.)			Closed
CAR 14	In Section 4.1: a) Please correct the version of AMS-III.D on pages 79 and 93. b) Please correct the notations (commas and dots) of the values in the table on page 80.	Section 4.1	a) Version of the applied methodology has been corrected on relevant pages. b) notations has been corrected in the relevant table.	a) Ok Closed (The versions were corrected.) b) Ok Closed (The notations were corrected.)			Closed

CAR 15	In Section 4.2: a) There is a contradiction for the value of Qbiogas,y on page 87. Please correct the contradiction. Also, in the table on page 87, the notation of the value of EFCH4 shall be corrected. b) Please correct the notations of the values in the table on page 89. c) Please correct the word "flares" on page 91. d) Please correct the version of Tool 06 on page 91. e) Please re-consider the sentence "Since the storage time of the manure after removal from the animal barns, including transportation, exceeds 24 hours before being fed into the anaerobic digester in the proposed project activity, PEstorage,y shall be accounted as zero." on page 91. f) For the calculation of PEtrans,y, please include whole calculation then simplify it in Section 4.2. g) Please re-	Section 4.2	a) Contradiction in the reported value of Qbiogas,y as well as notation has been corrected on page 87. b) Notations has been corrected. c) "flares statement on page 91 has been corrected as "flare". d) version of Tool06 has been corrected as 04.0. e) Related statement has been corredted as "Since the storage time of the manure after removal from the animal barns, including transportation, ***does not exceed 24 hours*** before being fed into the anaerobic digester in the proposed project activity, PEstorage,y shall be accounted as zero." f) Complete equation for the calculation of PEtrans,y as per AMS-III.AO has been provided and simplified in accordance with applied Methodology AMS-III.D. g) Qy value is reported in line with the methodological requirements.	a) Ok Closed (The contradiction and the notation were corrected.) b) Ok Closed (The notations were corrected.) c) Ok Closed (The relevant statement was corrected.) d) Ok Closed (The version of the tool was corrected.) e) Ok Closed (The relevant statement was corrected.) f) Ok Closed (The whole calculation was included.) g) Ok Closed (The clarification was made.)			Closed

	consider Qy value in Section 4.2.						
CAR 16	Please correct the notation of the value of Ru in Section 5.1.	Section 5.1	Notation of the Ru value has been corrected	Ok Closed (The notation was corrected.)			Closed
CAR 17	In Section 5.2: a) Please check the notations of the values in Section 5.2 (e.g. SVSLT,y). b) The version of "Methodological Tool: Project and leakage emissions from transportation of freight" is stated differently on page 110 compared to the other sections in the VCS PD. Please correct the contradiction. c) Please correct the value of the	Section 5.2	a) Notations has been corrected in Section 5.2. b) Relevant tool has been stated as "Methodological Tool "Project and leakage emissions from road transportation of freight" Version 1.1.0" throughout the PD. c) Value of mean annual precipitation has been corrected as 595.7 as per the meteorological data. d) The MCF value is reported in line with the "2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas	a) Ok Closed (The notations were corrected.) b) Ok Closed (The version was corrected throughout the PD.) c) Ok Closed (The value was corrected.) d) Ok Closed (The clarification was made.) e) Ok Closed (The value was			Closed

	mean annual precipitation on page 118. d) "Cool Temperate Dry" is stated for Yozgat province in Section 5.2. However, as per the relevant IPCC document, the value of MCF is not 0.67. Please check. e) Please indicate the value of EGy in Section 5.2. (on page 127).		Inventories (IPCC 2019), Volume 4, Chapter 10, Table 10.17 (Updated), page 10.68" as 0.67. e) EGy value has been indicated in Section 5.2.	included in Section 5.2.)			
NMF 1	N/A						
ADR 1	Please provide the first index protocol of the back-up electricity meter as well.	Section 1.11	First index protocol of the back-up electricity meter has been included in the relevant Supporting Document folder.	Ok Closed (The relevant document was provided.)			Closed
CL 1	Please indicate which technological challenges were faced in Section 1.8 with providing the relevant evidence documents.	Section 1.8	Relevant evidance has been provided as a supporting document.	Ok Closed (The evidence document was provided.)			Closed
CL 2	Please correct the numbering of the anaerobic digester tanks in Figure 1 in Section 1.11.	Section 1.11	Numbering of anaerobic digesters has been corrected for the figure in section 1.11.	Ok Closed (Figure 1 was revised accordingly.)			Closed
CL 3	Please include the contribution of SDG 8 in Section 1.17.	Section 1.17	contribution to SDG 8 has been included to Section 1.17.	Ok Closed (The contribution of SDG 8 was			Closed

				included in Section 1.17.)			
CL 4	Please specify the page number of the feasibility report which includes DM value.	Section 5.2	Page number for the DM value has been included to the relevent footnote on page 119 of the PD.	Ok Closed (The reference was included.)			Closed
FAR 1	The next VVB shall review the future changes in the manure supplier farm list.	Section 1.12					